Comparative Evaluation of 3D Hand Gesture Recognition Methods

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***Abstract*—This report presents a comparative study of six three-dimensional hand gesture recognition methods: Dynamic Time Warping (DTW), edit distance, Random Forest (RF), Decision Tree (DT), Logistic Regression (LR) and the $1 recognizer adapted to 3D. The experiments are conducted on two gesture domains from the dataset of Huang et al. [1]: domain 1 (Arabic numerals from 0 to 9) and domain 4 (procedural CAD symbols), each comprising ten gestures repeated ten times by ten users (i.e., 1 000 sequences per domain). All methods are evaluated according to two cross-validation protocols: user-independent (UI) and user-dependent (UD). An ablation study compares three preprocessing pipelines. Statistical significance is assessed by paired Wilcoxon signed-rank tests with Benjamini-Hochberg correction. Results show that standardisation drastically improves performance for most methods (gains up to +37.9 points), while PCA denoising degrades performance on both domains. On domain 1 (quasi-planar gestures), the $1 recognizer dominates in both UI (0.992 ± 0.011) and UD (0.999 ± 0.003) protocols, with statistically significant superiority over all other methods. On domain 4 (volumetric 3D gestures), DTW achieves the best performance in both UI (0.865 ± 0.078) and UD (0.987 ± 0.020) protocols, though differences are not statistically significant except for RF > DT. The user-dependent protocol systematically improves performance, with gains of +0.7 to +10.3 points on domain 1 and +7.0 to +15.2 points on domain 4, revealing that inter-user variability is the major bottleneck for complex 3D gestures.**

***Index Terms*—Gesture recognition, dynamic time warping, edit distance, random forest, $1 recognizer, 3D sequences, cross-validation, statistical testing.**

I Introduction

The rise of natural interfaces is one of the major challenges of contemporary human-computer interaction. Traditional interfaces based on the WIMP paradigm (windows, icons, menus and pointer) present limitations when one seeks to interact with three-dimensional environments. Gesture-based interfaces, exploiting the natural movement of the hand in space, constitute a promising alternative to this paradigm [1]. Today, depth sensors, widely available in devices made accessible to the general public, make it possible to record three-dimensional trajectories of the palm center. These are represented in the form of a sequence of position vectors $\mathbf{r}(t) = [x(t),\, y(t),\, z(t)]^\top$.

This report is part of the project for the course MLSM2154 – Artificial Intelligence, whose objective is to determine, through a comparative analysis, the best method(s) of statistical learning for supervised 3D gesture recognition. In total, six methods are investigated: two baseline methods whose foundations rest on dynamic programming (DTW and edit distance), three feature-based classifiers (Random Forest, Decision Tree, and Logistic Regression), and one template-matching method (the $1 algorithm adapted to 3D [2]). The evaluation of these models is conducted on two distinct gesture domains (Arabic numerals from 0 to 9 and procedural CAD symbols) in order to determine the generalisation capacity of the different methods.

This work seeks to answer various research questions, which are as follows. **RQ1:** What is the impact of preprocessing – standardisation and PCA denoising – on classification performance? **RQ2:** Which method achieves the best performance in user-dependent and user-independent modality? **RQ3:** Are the performance differences between methods statistically significant? **RQ4:** To what extent does the user-dependent protocol allow performance to be improved compared to the user-independent protocol?

The remainder of this report is organised as follows. Section II presents the detailed theoretical framework of each method, with their fundamental equations and associated intuitions. Section III describes the experimental protocol, that is, the datasets used and the associated statistical tests. Section IV presents and discusses the results obtained, answering in particular each of the research questions. Finally, Section V concludes by synthesising the main contributions and proposing directions for future work.

II Theoretical Background

II-A Notation and Common Background

Note: Throughout this work, the following notation conventions are adopted in a strictly consistent manner. A scalar is denoted in lowercase italic, e.g. t or n. A vector is denoted in lowercase bold, e.g. $\mathbf{r}(t) \in \mathbb{R}^3$. A matrix is denoted in uppercase bold, e.g. $\mathbf{W}$. A set is denoted in calligraphic font, e.g. $\mathcal{D}$.

As discussed in the introduction, a gesture is represented as a temporal sequence of T three-dimensional observations:

$$\mathbf{S} = \big(\mathbf{r}(1),\, \mathbf{r}(2),\, \ldots,\, \mathbf{r}(T)\big), \quad \mathbf{r}(t) = \begin{bmatrix} x(t) \\ y(t) \\ z(t) \end{bmatrix} \in \mathbb{R}^3 \tag{1}$$

where t denotes the discrete time-step index. Depending on the execution speed or the sensor sampling rate, two sequences from the same gesture may have different lengths T. It is precisely this temporal variability that justifies the use of alignment-robust methods such as DTW. The full dataset is denoted $\mathcal{D} = \{(\mathbf{S}_i, y_i)\}_{i=1}^{N}$, where $y_i \in \{0, \ldots, C-1\}$ denotes the class label of gesture i and N the total number of sequences.

II-B Preprocessing

To facilitate classification, three preprocessing steps are considered, corresponding to the three conditions of the ablation study: (a) no preprocessing, (b) per-gesture standardisation, (c) standardisation followed by PCA denoising.

II-B1 Per-Gesture Standardisation

Each gesture is recorded in a sensor-linked reference frame. The position of the palm center may vary depending on the user's posture in space, introducing scale and position biases across individuals. To neutralise these effects, each sequence is standardised independently, axis by axis. For a gesture $\mathbf{S}$ of length T, the standardised value of the k-th axis ($k \in \{x, y, z\}$) at time step t is defined by:

$$\tilde{r}_k(t) = \frac{r_k(t) - \mu_k}{\sigma_k}$$
$$\mu_k = \frac{1}{T}\sum_{t=1}^{T} r_k(t), \quad \sigma_k = \sqrt{\frac{1}{T}\sum_{t=1}^{T}\big(r_k(t) - \mu_k\big)^2} \tag{2}$$

If $\sigma_k = 0$ for a given axis, we conventionally set $\sigma_k = 1$ to avoid division by zero. The intuition behind this operation is straightforward: after standardisation, each gesture occupies a normalised space of zero mean and unit standard deviation on each axis. This removes inter-user amplitude differences without altering the shape of the trajectory. This step is a standard prerequisite in statistical learning when inputs are heterogeneous [3], [4].

II-B2 PCA Denoising (3D→2D→3D)

Depth sensors introduce non-negligible measurement noise, particularly on the axis perpendicular to the main plane of motion [1]. Indeed, for quasi-planar gestures (such as the Arabic numerals in domain 1), almost all the geometric variance is concentrated in a plane of space, and the third principal component largely captures noise.

Formally, let $\mathbf{X} \in \mathbb{R}^{T\times 3}$ be the matrix of T three-dimensional observations of a gesture after standardisation. Principal Component Analysis (PCA) factorises the empirical covariance matrix $\boldsymbol{\Sigma} = \frac{1}{T}\mathbf{X}^\top\mathbf{X}$ into its eigenvectors $\mathbf{V} = [\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3] \in \mathbb{R}^{3\times 3}$, ordered by decreasing variance. The projection onto the first two principal axes gives:

$$\mathbf{Z} = \mathbf{X}\,\mathbf{V}_{1:2} \in \mathbb{R}^{T\times 2} \tag{3}$$

where $\mathbf{V}_{1:2}$ denotes the first two columns of $\mathbf{V}$. The back-projection to the original 3D space is obtained by zeroing the third component before applying the inverse transform:

$$\widetilde{\mathbf{X}} = \mathbf{Z}\,\mathbf{V}_{1:2}^\top \in \mathbb{R}^{T\times 3} \tag{4}$$

The denoised sequence $\widetilde{\mathbf{X}}$ preserves the intrinsic planar geometry of the gesture while suppressing the variance carried by $\mathbf{v}_3$, interpreted as noise. An essential property of this approach is the absence of data leakage: the PCA is estimated on the T points of the gesture itself, without recourse to any external information [3]. The three explained variance ratios (EVR) — defined by $\mathrm{EVR}_j = \lambda_j / \sum_{k=1}^{3}\lambda_k$ where λ_j is the j-th eigenvalue — are retained as additional features for the random forest classifier in condition (c) of the ablation study, following the approach of Huang et al. [1].

II-B3 Vector Quantisation via k-Means

Edit distance operates on symbolic sequences, i.e. sequences of labels. It is therefore necessary to convert the continuous 3D trajectories into sequences of symbols (e.g. `AAACCCBAA...`). This conversion is achieved by vector quantisation (clustering) using the k-means algorithm [3].

Let k be the number of centroids. At each fold of cross-validation, the k-means algorithm is fitted exclusively on the set of 3D points from the training sequences. For a given fold, the training set contains N_{train} sequences, and all their $\sum_i T_i$ 3D points are gathered into a single point cloud on which the k centroids $\{\mathbf{c}_1, \ldots, \mathbf{c}_k\} \subset \mathbb{R}^3$ are estimated. Each observation $\mathbf{r}(t)$ of a sequence — whether belonging to the train or test set — is then replaced by the label of the nearest centroid using Euclidean distance:

$$l(t) = \arg\min_{j\in\{1,\ldots,k\}} \|\mathbf{r}(t) - \mathbf{c}_j\|_2 \tag{5}$$

The continuous sequence is thus transformed into the symbolic sequence $(l(1), l(2), \ldots, l(T)) \in \{1, \ldots, k\}^T$, exploitable by edit distance. Performing this clustering *inside* cross-validation, on the training set alone, is the most rigorous approach: it guarantees that the centroids are never influenced by test data, in accordance with the course guidelines [5].

II-C Baseline Methods

II-C1 Dynamic Time Warping (DTW)

Dynamic Time Warping is an algorithm for measuring the similarity between two temporal sequences, initially proposed in the context of speech recognition [6], and subsequently used to compare multidimensional time series [3]. Its fundamental principle consists in finding the optimal temporal alignment between two sequences of potentially different lengths, by allowing local elongations and compressions.

Let $\mathbf{S}_1 = (\mathbf{r}_1(1), \ldots, \mathbf{r}_1(n))$ and $\mathbf{S}_2 = (\mathbf{r}_2(1), \ldots, \mathbf{r}_2(m))$ be two sequences. The local distance between time steps i and j is defined by the Euclidean distance between the corresponding observations:

$$d(i,j) = \|\mathbf{r}_1(i) - \mathbf{r}_2(j)\|_2 \tag{6}$$

The DTW distance is then computed by dynamic programming via the following recurrence on an accumulation matrix $\mathbf{G} \in \mathbb{R}^{(n+1)\times(m+1)}$, initialised with $g(0,0) = 0$ and $g(i,0) = g(0,j) = +\infty$ for $i, j > 0$:

$$g(i,j) = \min\begin{cases} g(i-1,\,j) + d(i,j) \\ g(i-1,\,j-1) + 2\cdot d(i,j) \\ g(i,\,j-1) + d(i,j) \end{cases} \tag{7}$$

where the vertical and horizontal transitions have a unit cost $d(i, j)$, while the diagonal transition is weighted by a factor of 2 in order to penalise successive repetitions and to balance the different types of alignment [6]. The value $g(n, m)$ gives the minimum cumulative cost of the optimal warping path. Following standard practice [6], this value is normalised by the sum of the lengths in order to make the distance comparable across pairs of sequences of different lengths:

$$d_{\mathrm{DTW}}(\mathbf{S}_1, \mathbf{S}_2) = \frac{g(n, m)}{n + m} \tag{8}$$

Intuition. Where a rigid point-to-point distance would severely penalise a gesture executed more slowly, DTW "stretches" or "compresses" the temporal axis locally to minimise misalignment. This is particularly relevant in our context, where sequence lengths vary considerably (from a few tens to several hundreds of time steps depending on the user).

II-C2 Edit Distance

Edit distance, also known as Levenshtein distance, quantifies the minimum number of elementary operations required to transform one symbolic sequence into another [3], [7]. The three allowed operations are the insertion of a symbol, the deletion of a symbol, and the substitution of one symbol by another, each with unit cost.

Let $\mathbf{l}^1 = (l_1^1, \ldots, l_{L_1}^1)$ and $\mathbf{l}^2 = (l_1^2, \ldots, l_{L_2}^2)$ be two label sequences obtained from k-means vector quantisation. Edit distance is computed by dynamic programming via the matrix $\mathbf{M} \in \mathbb{N}^{(L_1+1)\times(L_2+1)}$, initialised by $m(i, 0) = i$ and $m(0, j) = j$, then filled by:

$$m(i, j) = \min \begin{cases} m(i-1, j) + 1 & \text{(deletion)} \\ m(i, j-1) + 1 & \text{(insertion)} \\ m(i-1, j-1) + \delta(i, j) & \text{(substitution)} \end{cases} \tag{9}$$

where $\delta(i, j) = 0$ if $l_i^1 = l_j^2$ and $\delta(i, j) = 1$ otherwise. The final distance is $d_{\mathrm{edit}}(\mathbf{l}^1, \mathbf{l}^2) = m(L_1, L_2)$.

Intuition. Edit distance captures the similarity between two trajectories through their symbolic representation. Two similar gestures will produce close label sequences, requiring few transformation operations.

II-D Advanced Methods

II-D1 Random Forest

Random Forest (RF) is an ensemble learning method introduced by Breiman [8] and based on the principle of bagging (bootstrap aggregating). An ensemble of B decision trees $\{h_b(\mathbf{f})\}_{b=1}^{B}$ is constructed, each trained on a subsample drawn with replacement from the training set. At each node of each tree, the best split is searched not over the entirety of the p features, but over a random subset of size $\lfloor\sqrt{p}\rfloor$, which introduces decorrelation between the trees and reduces the global variance. The final prediction is obtained by majority voting:

$$\hat{y} = \arg\max_{c} \sum_{b=1}^{B} \mathbf{1}[h_b(\mathbf{f}) = c] \tag{10}$$

In our implementation, $B = 200$ trees are used with `max_features` $= \sqrt{p}$, values justified by the recommendations of Breiman [8] to obtain a low generalisation variance.

a) Feature Extraction

Unlike DTW and edit distance which operate directly on temporal sequences, Random Forest requires a fixed-dimension feature vector. For each gesture $\mathbf{S} = \{\mathbf{r}(1), \ldots, \mathbf{r}(T)\}$, we extract $p = 44$ features (or $p = 47$ if the three explained variance ratios from PCA are included), distributed across nine categories: per-axis statistics (21 features: mean, standard deviation, min, max, range, skewness, kurtosis for x, y, z), instantaneous velocity (4 features: mean, standard deviation, max, min), acceleration (2 features: mean norm and standard deviation), total arc length (1 feature), bounding box (4 features: extents per axis + diagonal), curvature via angles between successive velocity vectors (3 features: mean, standard deviation, sum), velocities and displacements per temporal segment (3 + 3 features for the beginning/middle/end thirds), absolute variations per axis (3 features), and conditional PCA explained variance ratios (3 features if condition (c) applied).

b) Feature Selection and Optimisation

Feature selection is performed at each cross-validation fold, exclusively on the training set [9], via a three-stage pipeline: (1) variance filter (removal of quasi-constant features, variance $< 10^{-10}$), (2) correlation-based redundancy removal (Pearson $|r| > 0.99$, Yu & Liu [10]), (3) permutation importance on 80/20 split to retain features covering 95% of cumulative importance [8], [11].

Hyperparameters are selected at each fold via GridSearchCV (5 inner folds, Varma & Simon [12]) on the grid: `n_estimators` $\in \{100, 200\}$, `max_depth` $\in \{5, 10, 15\}$, `min_samples_split` $\in \{5, 10\}$, `min_samples_leaf` $\in \{2, 5\}$.

II-D2 Decision Tree

Decision Tree (DT) is a supervised non-parametric classifier that recursively partitions the feature space via binary tests on features [13], [14].

Its inclusion in our study serves a pedagogical comparison objective: confronting a single tree (DT) with an ensemble of trees (Random Forest) allows us to empirically demonstrate the contribution of bagging on this dataset [8], [15]. A single tree is subject to overfitting (high variance), while Random Forest reduces this variance through aggregation.

DT uses the same 44 features (or 47 with EVR) as Random Forest, with the same three-stage selection pipeline described in Section II-D1. Hyperparameters are selected at each fold via GridSearchCV (5 folds) on the grid: `max_depth` $\in \{5, 10, 15\}$, `min_samples_split` $\in \{5, 10, 20\}$, `min_samples_leaf` $\in \{2, 5, 10\}$, and `criterion` $\in$ {"gini", "entropy"}.

II-D3 Logistic Regression

Multinomial Logistic Regression (LR) is a linear model for multi-class classification [16], [17]. Weights are estimated by minimising cross-entropy with an L2 penalty (ridge regularisation) controlled by the hyperparameter C, via the lbfgs algorithm with `max_iter` $= 5000$.

Its role in this study is to serve as a linear baseline: as a linear model, LR provides a lower performance bound against which non-linear methods (RF, DT) and temporal alignment-based methods (DTW, edit distance, \$1) can be compared. If LR obtains performances comparable to non-linear methods, this suggests that the task is linearly separable; otherwise, it justifies the use of more complex models.

LR systematically applies internal standardisation (StandardScaler) before weight estimation, which makes preprocessing conditions (a) and (b) functionally equivalent for this classifier. LR uses the same 44 features (or 47 with EVR) as RF and DT, with the same selection pipeline. Hyperparameter C is selected via GridSearchCV (5 folds) on the grid $C \in \{0.01, 0.1, 1, 10, 100\}$.

II-D4 $1 Recognizer — 3D Adaptation

The \$1 algorithm by Wobbrock, Wilson and Li [2] was initially designed for 2D gesture recognition with minimal preprocessing. We adopt here the canonical 3D extension proposed by Kratz & Rohs [18], entitled "A \$3 Gesture Recognizer".

a) Preprocessing Pipeline

Each gesture $\mathbf{S} = \{\mathbf{r}(1), \ldots, \mathbf{r}(T)\}$ is normalised according to four sequential steps:

1) **Resampling**: The sequence is resampled to $N = 150$ equidistant points along its arc length, ensuring uniform temporal length.
2) **Rotation to indicative reference frame**: The vector from centroid $\mathbf{c}$ to the first point $\tilde{\mathbf{p}}_1$ defines a reference direction. Rotation is applied via Rodrigues' formula ($\mathbf{r}_{\text{rot}} = \mathbf{r}\cos(\theta) + (\mathbf{k} \times \mathbf{r})\sin(\theta) + \mathbf{k}(\mathbf{k} \cdot \mathbf{r})(1 - \cos(\theta))$ where $\mathbf{k}$ is the normalised rotation axis) to align all gestures according to a standard orientation. This step must be performed before translation to the origin.
3) **Translation to origin**: The gesture is translated so that its centroid coincides with the origin.
4) **Uniform scaling**: The gesture is scaled so that its largest dimension equals $l = 1.0$. This uniform scaling (not axis-by-axis) preserves spatial proportions and avoids division by zero for quasi-planar gestures.

b) Recognition and Score

To classify a candidate gesture, it is preprocessed then compared to the set of preprocessed templates via Golden Section Search (GSS). GSS is performed sequentially on three rotation axes (x, y, z) to find the optimal angle minimising the point-to-point MSE distance:

$$d(\mathbf{S}_{\text{cand}}, \mathbf{S}^{\text{template}}) = \frac{1}{N} \sum_{j} \|\mathbf{r}_{\text{cand}}(j) - \mathbf{r}^{\text{template}}(j)\|_2^2 \tag{11}$$

GSS uses the golden ratio $\phi = 0.5(\sqrt{5} - 1)$, with a convergence threshold of 2 degrees and 11 iterations per axis. The recognition score is calculated by:

$$S = 1 - \frac{d}{0.5 \cdot \sqrt{3} \cdot l^2} \tag{12}$$

where the denominator represents the theoretical upper bound of the MSE distance. The final prediction is obtained by 1-NN (the closest template). An optional rejection heuristic exists (threshold $\epsilon = 0.60$) but is disabled in our evaluation (`allow_rejection = False`) to ensure comparability with other methods.

Essential property: Templates are preprocessed only once and cached [2], significantly reducing computational cost.

II-E Classification: k-Nearest Neighbours (k-NN)

Unlike RF, DT and LR, the DTW, edit distance and \$1 recognizer methods do not directly produce a class label. They compute a dissimilarity measure between two sequences. Classification is therefore performed by the k-nearest-neighbours (k-NN) classifier with $k = 1$ [3]. For a test gesture $\mathbf{S}_{\text{test}}$, the predicted class is the class of the training gesture most similar according to the retained dissimilarity measure:

$$\hat{y} = y_{\arg\min_{i \in \mathcal{T}} d(\mathbf{S}_{\text{test}}, \mathbf{S}_i)} \tag{13}$$

where $\mathcal{T}$ denotes the training set. In user-independent modality, $\mathcal{T}$ contains the set of sequences from the nine training users. In user-dependent modality, the search for the nearest neighbour is restricted to the sequences of the same user as the test gesture: the model is thus personalised to each individual.

The choice $k = 1$ (1-NN) is motivated by the gesture recognition literature, where 1-NN constitutes the canonical non-parametric baseline [2], [19]–[21]. 1-NN provides a transparent baseline without hyperparameters.

III Experimental Setup

III-A Dataset

The two datasets used in this study come from the study of Huang et al. [1]. Each domain comprises ten gesture classes, recorded by ten users each performing ten repetitions per class, corresponding to $N = 1000$ sequences per domain. The spatial coordinates are recorded using a depth-sensing camera (SoftKinetic DepthSense), and time steps are considered equidistant, in accordance with the project guidelines. The completeness of the dataset was verified: the 1000 expected sequences in each domain are present, with no missing data.

Domain 1 contains Arabic numerals from 0 to 9 drawn in the air. For this domain, gestures are considered quasi-planar, making it a moderately difficult domain. Domain 4, on the other hand, contains ten procedural CAD symbols (Cuboid, Cylinder, Sphere, Rectangular Pipe, Hemisphere, Cylindrical Pipe, Pyramid, Tetrahedron, Cone, Toroid) of a three-dimensional nature, naturally making the classification task more demanding. A visual representation of a representative sample per class, obtained by projection onto the XY plane, is presented in Figure 1. Furthermore, the descriptive characteristics of both domains are summarised in Table 1.

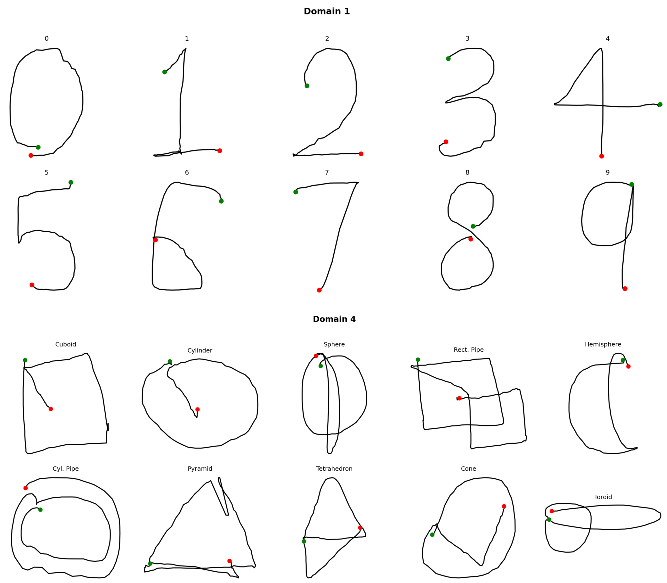

Figure 1: *Representative sample per gesture class projected onto the XY plane. Top: Domain 1 (Arabic numerals 0–9). Bottom: Domain 4 (CAD symbols). Green dots indicate the start of the trajectory; red dots indicate the end.*

	Domain 1	**Domain 4**
Gesture classes	0–9 (Arabic numerals)	1–10 (CAD symbols)
Number of classes	10	10
Number of users	10	10
Rep. per class per user	10	10
Total sequences	1 000	1 000
Min. length (steps)	31	56
Max. length (steps)	240	314
Mean length (steps)	85.1 ± 26.5	139.6 ± 43.7
Dominant geometry	Quasi-planar	Three-dimensional

Table 1: *Descriptive characteristics of the two gesture domains.*

III-B Exploratory Analysis

A preliminary exploratory analysis is conducted in order to motivate the methodological choices made in this work. It focuses on two complementary aspects: the distribution of sequence lengths and the geometric nature of the trajectories.

The box plots of lengths per class (Figures 6 and 7, Appendix) reveal notable variability within each domain. For domain 1, the medians range between 50 and 115 time steps depending on the class, with extreme values reaching 240 steps. Class 7 presents the lowest median, while class 8 is the longest. For domain 4, the variability is even more pronounced. Indeed, the medians range from approximately 105 to 215 steps, with extreme values reaching 314 steps. Class 4 (Rectangular Pipe) notably presents a particularly large interquartile dispersion, reflecting strong inter-user heterogeneity in the execution of this gesture. This variability renders any point-to-point comparison between sequences unsuitable, and constitutes the justification for the use of DTW and edit distance. These two methods are precisely designed to compare sequences of different lengths [6].

The gesture trajectories highlight a structural difference between the two domains. For domain 1, the trajectories are contained within a plane in space: the z-axis presents a low amplitude of variation. This quasi-planar nature is consistent with the fact that digits are drawn in a frontal plane. For domain 4, however, the gestures fully exploit all three dimensions of space, explained by the volumetric nature of CAD objects. This observation motivates the application of PCA denoising (Section II). For domain 1, the third principal component is expected to carry noise, and its elimination should improve performance. In contrast, for domain 4, this same component carries real geometric information, and denoising could prove detrimental. The ablation study (Section III-D) will quantify these effects.

III-C Cross-Validation Protocol

Two cross-validation protocols are implemented, in accordance with the project guidelines.

User-independent protocol (UI). A leave-one-user-out procedure with 10 folds is applied. At each fold $f \in \{1, \ldots, 10\}$, the test set contains all sequences from user f (i.e., 100 sequences), and the training set contains the sequences from the nine remaining users (i.e., 900 sequences). This protocol aims to evaluate the generalisation capacity of the model to a user unknown at training time. This constitutes the most demanding condition for real-world deployment.

User-dependent protocol (UD). A leave-one-sample-out procedure with 10 folds is applied. At each fold f, the f-th repetition of each gesture from each user is removed from training and placed in the test set. The test set thus contains 100 sequences and the training set contains 900. In UD mode, the nearest-neighbour search is restricted to sequences from the same user as the test gesture (in accordance with the provided instructions). This allows the capacity of the system to recognise a new gesture from a specific user known to the model to be evaluated. It is naturally expected that UD performance will be superior to UI performance, given that inter-individual variability no longer penalises classification (RQ4).

Fundamental property: shared fold indices. The fold indices are generated once and shared across all methods. This decision ensures a coherent basis for comparison. By fixing these indices in advance, one ensures that performance differences between models are not due to a random partition of the data, but to the actual capabilities of the algorithms [3].

III-D Ablation Study

In order to study the impact of each preprocessing step on classification performance (RQ1), an ablation study is conducted in user-independent and user-dependent modalities, across all methods and domains. Three preprocessing conditions are evaluated: (a) no preprocessing, (b) per-gesture and per-axis sequence standardisation, (c) standardisation + PCA denoising. The optimal preprocessing may differ according to the protocol: for example, PCA may benefit the UI protocol (inter-user) more than the UD protocol (intra-user). For each method and each domain, the condition having produced the highest mean accuracy over the 10 folds is selected separately for UI and UD, guaranteeing that each method is evaluated under its best preprocessing conditions for each protocol.

III-E Hyperparameter Selection

The optimal number of centroids k for vector quantisation (Section II-B3) is determined via empirical validation curves exploring $k \in \{5, 10, 15, 20, 25, 30, 40\}$ in user-independent cross-validation on standardised data [22] (Figures 8 and 9, Appendix). The values maximising mean UI accuracy are retained: $k = 20$ for Domain 1 and $k = 15$ for Domain 4. These values correspond to the optimal preprocessing condition (b) — standardisation alone — which is retained for edit distance throughout the main evaluation.

Similarly, a k-NN validation curve exploring $k \in \{1, 3, 5, 7, 9\}$ is produced for transparency (Figures 10 and 11, Appendix). While Domain 1 achieves its optimum at $k = 1$ (0.918), Domain 4 shows a marginal improvement at $k = 3$ (0.867 vs 0.865, a gain of only +0.002). Given this negligible difference and the standard practice in gesture recognition literature [2], [19]–[21], $k = 1$ is enforced throughout the pipeline for both domains (Section II-E), providing a transparent, parameter-free baseline that avoids the risk of overfitting through majority voting with small k.

The hyperparameters of RF, DT and LR are selected per fold via GridSearchCV (5 inner folds), while those of DTW and edit distance are fixed once for all folds. This asymmetry is justified by the nature of the hyperparameter spaces: RF, DT, and LR have combinatorial grids requiring nested cross-validation to avoid overfitting the hyperparameter selection itself. In contrast, DTW and edit distance depend on a single scalar hyperparameter exhibiting a clear plateau in validation curves, making a single global selection both safe and computationally efficient.

III-F Overfitting Diagnostic

To evaluate the risk of overfitting for parametric classifiers (RF, DT, LR), the gap between training accuracy and test accuracy is calculated for each fold. A large gap indicates high variance. DTW, edit distance and $1 are excluded because their training accuracy is 1.0 by construction (1-NN).

III-G Statistical Tests

In order to evaluate the statistical significance of performance differences between methods (RQ3), paired Wilcoxon signed-rank tests are performed on the user-independent results of each domain.

Per-(gesture, user) accuracy aggregation: For each method, rather than directly comparing the 10 fold accuracies, we construct a vector of $n = 100$ per-(gesture, user) accuracies, ordered by (gesture ascending, user ascending). This granularity ensures greater statistical power by testing on 100 paired observations rather than 10.

Pairwise comparisons: For each of the 15 method pairs (6 methods), a paired Wilcoxon signed-rank test is performed. The null hypothesis is that the two methods have identical performance distributions.

Multiple testing correction: In order to control Type I error inflation due to multiple comparisons, the Benjamini-Hochberg (BH) False Discovery Rate correction is applied [23]. BH is appropriate here because the tests share methods across pairs and are therefore not independent—BH is the correct procedure under positive dependence. Statistical significance is declared at $\alpha = 0.05$ after BH correction.

IV Results and Discussion

IV-A Ablation Study: Impact of Preprocessing (RQ1)

The ablation study evaluates the impact of three preprocessing pipelines on classification performance: (a) no preprocessing, (b) standardisation alone, and (c) standardisation followed by PCA denoising. Results are presented separately for user-independent (UI) and user-dependent (UD) protocols on both domains.

IV-A1 Domain 1: Arabic Numerals (Quasi-Planar Gestures)

User-Independent Protocol (UI). Table 2 synthesises the obtained results. Standardisation (condition b) drastically improves performance for all methods except the $1 recognizer. For DTW, the gain is +29.4 points (from 0.624 to 0.918), for Edit Distance +31.7 points (from 0.552 to 0.869), and for RF +12.1 points (from 0.791 to 0.912). The observed improvements confirm that standardisation effectively neutralises inter-user scale differences, in accordance with theoretical expectations [3]. The $1 recognizer, on the other hand, performs better without preprocessing (0.992) than with standardisation (0.885), because its internal pipeline integrating normalisation (uniform scaling to a cube of size $l = 1.0$) is already robust to scale variations [2].

Method	(a) Raw	(b) Std	(c) Std+PCA	Best
Edit Distance	0.552 ± 0.189	**0.869 ± 0.138**	0.828 ± 0.140	(b)
DTW	0.624 ± 0.167	**0.918 ± 0.097**	0.879 ± 0.108	(b)
DT	0.685 ± 0.068	**0.827 ± 0.081**	0.682 ± 0.111	(b)
RF	0.791 ± 0.095	**0.912 ± 0.055**	0.808 ± 0.059	(b)
LR	0.864 ± 0.171	**0.870 ± 0.093**	0.823 ± 0.112	(b)
$1	**0.992 ± 0.011**	0.885 ± 0.089	0.860 ± 0.080	(a)

Table 2: *Ablation study — Domain 1, User-independent. Best condition per method in bold.*

Adding PCA denoising to standardisation (condition c) slightly degrades performance compared to standardisation alone. Indeed, DTW drops from 0.918 to 0.879, RF from 0.912 to 0.808, DT from 0.827 to 0.682. This degradation was expected for Domain 1. Although gestures are quasi-planar, the analysis of EVR distributions (Figure 2) reveals that PC3 captures on average 5.8% of the total variance. If PC3 contained only noise, PCA projection could only benefit performance. The observed degradation suggests that PC3, even weak, encodes useful geometric information.

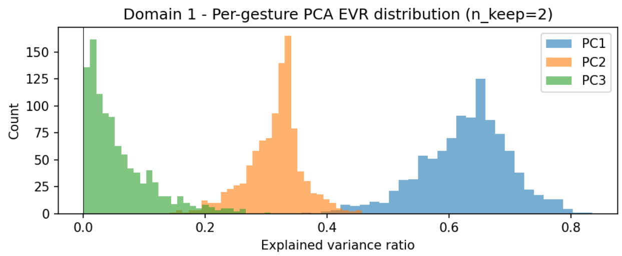

Figure 2: *Per-gesture PCA explained variance ratio (EVR) distribution for Domain 1.*

User-Dependent Protocol (UD). Trends are similar, with overall higher performance (Table 3). Standardisation remains beneficial for DTW (+1.0 point), Edit Distance (+0.8 point), DT (+5.0 points), RF (+0.7 point), and LR (+0.1 point). The $1 recognizer achieves quasi-perfect accuracy without preprocessing (0.999 ± 0.003), confirming that its internal pipeline suffices for this domain. PCA denoising again degrades performance for all models.

Method	(a) Raw	(b) Std	(c) Std+PCA	Best
Edit Distance	0.964 ± 0.031	**0.972 ± 0.022**	0.972 ± 0.019	(b)
DTW	0.978 ± 0.022	**0.988 ± 0.010**	0.967 ± 0.013	(b)
DT	0.825 ± 0.077	**0.875 ± 0.049**	0.766 ± 0.074	(b)
RF	0.949 ± 0.045	**0.956 ± 0.035**	0.888 ± 0.094	(b)
LR	0.955 ± 0.037	**0.956 ± 0.037**	0.917 ± 0.061	(b)
$1	**0.999 ± 0.003**	0.962 ± 0.013	0.935 ± 0.019	(a)

Table 3: *Ablation study — Domain 1, User-dependent. Best condition per method in bold.*

IV-A2 Domain 4: CAD 3D Symbols (Volumetric Gestures)

User-Independent Protocol (UI). Results (Table 4) show that standardisation is even more critical for Domain 4 than for Domain 1. DTW gains +37.9 points (from 0.486 to 0.865), Edit Distance +34.5 points (from 0.492 to 0.837), and RF +3.7 points (from 0.827 to 0.864). These massive gains are explained by the 3D nature of gestures: scale and position variations are naturally more pronounced in a volumetric space, and standardisation neutralises them effectively.

Method	(a) Raw	(b) Std	(c) Std+PCA	Best
Edit Distance	0.492 ± 0.149	**0.837 ± 0.090**	0.702 ± 0.085	(b)
DTW	0.486 ± 0.117	**0.865 ± 0.078**	0.752 ± 0.045	(b)
DT	0.692 ± 0.097	**0.813 ± 0.072**	0.573 ± 0.066	(b)
RF	0.827 ± 0.083	**0.864 ± 0.088**	0.718 ± 0.067	(b)
LR	**0.851 ± 0.105**	0.825 ± 0.112	0.708 ± 0.062	(a)
$1	0.821 ± 0.085	**0.831 ± 0.063**	0.705 ± 0.038	(b)

Table 4: *Ablation study — Domain 4, User-independent. Best condition per method in bold.*

EVR analysis (Figure 3) reveals that PC3 captures on average 15.7% of the total variance — significantly more than the 5.8% of Domain 1. This PC3 encodes real 3D geometric information, essential for distinguishing volumetric symbols such as cylinder, sphere, or cone. Removing PC3 therefore eliminates a discriminant dimension.

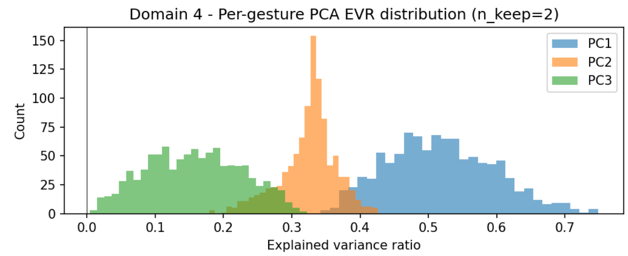

Figure 3: *Per-gesture PCA explained variance ratio (EVR) distribution for Domain 4.*

Logistic Regression presents atypical behaviour: it performs better without preprocessing (0.851) than with standardisation (0.825) or standardisation + PCA (0.708). This counter-intuitive result may be explained by an interaction between LR's internal standardisation (systematic StandardScaler before optimisation) and the extracted features. On raw data, certain high-magnitude features (such as `arc_length` or `bbox_diag`) naturally dominate the weight vector, effectively capturing volumetric geometry. External standardisation artificially equalises all features, which can dilute the importance of key geometric features and degrade linear separability [17].

User-Dependent Protocol (UD). Performance improves overall (Table 5), with trends similar to the UI protocol. Standardisation remains beneficial for all methods except $1. PCA denoising again systematically degrades all performances. This degradation confirms that PC3 encodes discriminant information for 3D gestures, and that its removal harms classification.

Method	(a) Raw	(b) Std	(c) Std+PCA	Best
Edit Distance	0.945 ± 0.032	**0.984 ± 0.017**	0.949 ± 0.031	(b)
DTW	0.952 ± 0.029	**0.987 ± 0.020**	0.952 ± 0.028	(b)
DT	0.863 ± 0.041	**0.883 ± 0.026**	0.745 ± 0.072	(b)
RF	0.958 ± 0.025	**0.976 ± 0.021**	0.901 ± 0.048	(b)
LR	0.952 ± 0.030	**0.968 ± 0.018**	0.924 ± 0.040	(b)
$1	**0.973 ± 0.018**	0.956 ± 0.024	0.913 ± 0.029	(a)

Table 5: *Ablation study — Domain 4, User-dependent. Best condition per method in bold.*

IV-A3 RQ1 Summary: Impact of Preprocessing

Standardisation drastically improves performance for DTW, Edit Distance, and feature-based classifiers (RF, DT, LR), with gains reaching +37.9 points on Domain 4. It is critical for neutralising inter-user scale and position variations. PCA denoising, on the other hand, degrades performance on both domains, including on the quasi-planar Domain 1 where it was expected to be beneficial. This degradation reveals that PC3, even weak (5.8% variance on D1), encodes discriminant geometric information that is lost during 2D projection [3], [4]. The $1 recognizer generally performs better without external preprocessing thanks to its robust internal normalisation pipeline. The sole exception is Domain 4 UI, where standardisation provides a marginal improvement (0.831 vs 0.821, +1 point), though performance remains comparable (for this reason, subsequent results consider $1 with raw data for methodological consistency across all domains and protocols). The optimal condition is therefore (b) standardisation alone for 5/6 methods, and (a) raw data for $1.

IV-B Main Classification Results (RQ2, RQ4)

Table 6 synthesises the final performance of each method, evaluated under its optimal preprocessing condition (determined in Section 4.A).

Dom.	Prot.	$1	DT	DTW	ED	LR	RF
1	UI	**0.992** ±0.011	0.827 ±0.081	0.918 ±0.097	0.869 ±0.138	0.870 ±0.093	0.912 ±0.055
	UD	**0.999** ±0.003	0.875 ±0.049	0.988 ±0.010	0.972 ±0.022	0.956 ±0.037	0.956 ±0.035
4	UI	0.821 ±0.085	0.813 ±0.072	**0.865** ±0.078	0.837 ±0.090	0.851 ±0.105	0.864 ±0.088
	UD	0.973 ±0.018	0.883 ±0.026	**0.987** ±0.020	0.984 ±0.017	0.968 ±0.018	0.976 ±0.021

Table 6: *Final performance (mean accuracy ± standard deviation) under optimal preprocessing condition per method. Best performance per row in bold. Dom. = Domain, Prot. = Protocol.*

IV-B1 Domain 1: Domination of the $1 Recognizer

UI Protocol. The $1 recognizer dominates widely with an accuracy of 0.992 ± 0.011, surpassing all other methods by +7.4 to +16.5 points. Temporal alignment-based methods (DTW: 0.918, Edit Distance: 0.869) obtain good performance, confirming their robustness to sequence length variations. Random Forest (0.912) and Logistic Regression (0.870) perform honourably, demonstrating that the 44 extracted features effectively capture the geometry of quasi-planar digits. Decision Tree (0.827) presents the weakest performance among evaluated methods, probably suffering from overfitting despite hyperparameter selection via GridSearchCV.

The superiority of $1 on Domain 1 is explained by several factors: (1) Arabic numerals drawn in the air are quasi-planar trajectories with strong intra-class shape consistency. The $1 normalisation pipeline (resampling $N = 150$, rotation to indicative reference frame, uniform scaling) perfectly preserves these shapes. (2) Golden Section Search on the rotation axis optimises angular alignment, capturing micro-variations in orientation that distinguish geometrically close digits (such as 6 and 9). (3) 1-NN with point-to-point MSE distance benefits from a well-structured template space, where digits are naturally separable after normalisation.

UD Protocol. The $1 achieves quasi-perfect accuracy of 0.999 ± 0.003, with only 1 error out of 1000 tested sequences. This result, although impressive, is consistent with performances reported by Wobbrock et al. [2], who observe recognition rates exceeding 99% for simple gestures in user-dependent mode on their dataset of 16 unistroke gestures. The authors explain that the $1 normalisation pipeline (resampling, indicative rotation, uniform scaling) systematically eliminates variations in position, orientation and scale, making geometric shapes directly comparable. When inter-user variability is suppressed (UD protocol), this normalisation allows quasi-perfect class separation. In our case, quasi-planar Arabic numerals fully benefit from this geometric normalisation.

DTW (0.988) and Edit Distance (0.972) also improve significantly, proving that by comparing a user only to themselves, natural differences between individuals are eliminated. RF and LR both reach 0.956, demonstrating that extracted features generalise well intra-user. DT remains the weakest at 0.875, suggesting persistent overfitting despite the UD protocol.

IV-B2 Domain 4: Balance of Methods

UI Protocol. Unlike Domain 1, no method clearly dominates. DTW (0.865) obtains the best performance, closely followed by RF (0.864) and LR (0.851). The $1 recognizer drops drastically to 0.821 (-17.1 points compared to D1), revealing its limitations on volumetric 3D gestures.

DTW performs remarkably well (0.865). Its dynamic recurrence tolerates temporal elongations and compressions, effectively capturing execution variations of 3D symbols. Random Forest and Logistic Regression (0.864 and 0.851) demonstrate that extracted features encode sufficient discriminant information to separate volumetric symbols.

UD Protocol. Performance improves drastically for all methods. DTW reaches 0.987 (+12.2 points), Edit Distance 0.984 (+14.7 points), RF 0.976 (+11.2 points), and $1 0.973 (+15.2 points). This massive improvement confirms that inter-user variability is the major limiting factor on Domain 4. By restricting k-NN to the same user, the UD protocol eliminates differences in execution style (speed, amplitude, spatial orientation).

IV-C Comparative Analysis: Domain 1 vs Domain 4

Relative difficulty. Domain 4 (3D CAD symbols) is significantly more difficult than Domain 1 (quasi-planar Arabic numerals). In UI protocol, the best method on D1 ($1: 0.992) surpasses the best on D4 (DTW: 0.865) by +12.7 points. This difference is explained by: (1) Intrinsic dimensionality: digits are quasi-planar (PC1+PC2 capture 94.2% of variance), while CAD symbols fully exploit 3D space (PC3 retains 15.7% of variance). (2) Geometric complexity: CAD symbols (cylinder, sphere, cone) present complex volumetric shapes increasing inter-class confusion.

Answer to RQ2: Domain 1: the $1 recognizer is best in UI (0.992) and UD (0.999), thanks to its normalisation pipeline optimised for planar gestures. **Domain 4:** DTW is best in

UI (0.865) and UD (0.987), demonstrating its robustness to complex 3D gestures. RF and LR perform honourably on both domains, confirming that extracted features effectively capture gestural geometry.

Answer to RQ4: The UD protocol systematically improves performance on both domains. On D1, UD→UI gains are moderate (+0.7 to +10.3 points), as digits are already well discriminable. On D4, gains are massive (+7.0 to +15.2 points), revealing that inter-user variability is the main bottleneck for complex 3D gestures. The UD protocol eliminates this variability by restricting k-NN to the same user, allowing methods to achieve quasi-perfect performance.

IV-D Confusion Matrix Analysis

Confusion matrices of the best models ($1 on Domain 1 UI; DTW on Domain 4 UI) reveal inter-class confusions and geometric limitations of each method.

IV-D1 Domain 1: $1 Recognizer (UI)

The matrix (Figure 4) is quasi-diagonal, confirming the excellence of $1. Observed errors are: (1) Class 1 → 2: 1 error. (2) Class 2 → 1: 4 errors. (3) Class 5 → 1: 2 errors. (4) Class 9 → 7: 1 error. These confusions are geometrically justified: digits 1 and 2 share an initial vertical trajectory, and execution variations (insufficiently pronounced curvature of "2") can make discrimination difficult. Confusions 5→1 and 9→7 are more surprising, suggesting potentially atypical trajectories in the test set.

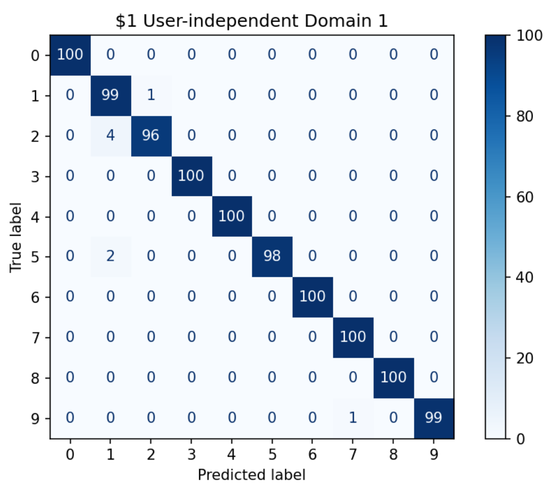

Figure 4: *Confusion matrix for $1 recognizer on Domain 1 (User-independent).*

IV-D2 Domain 4: DTW (UI)

The matrix (Figure 5) reveals more marked confusions: (1) Class 2 (Cylinder) → 1 (Cuboid): 39 errors (39%). Cylinder and cuboid share a basic rectangular structure, and DTW aligns their sequences similarly. (2) Class 3 (Sphere) → Multiple: 30% of errors dispersed. The sphere, a closed circular trajectory, is confused with symbols presenting curved components (cylinder, hemisphere). (3) Class 8 (Tetrahedron) → 7 (Cylindrical Pipe): 16 errors (16%). Both symbols share similar trajectory patterns, making discrimination difficult for DTW.

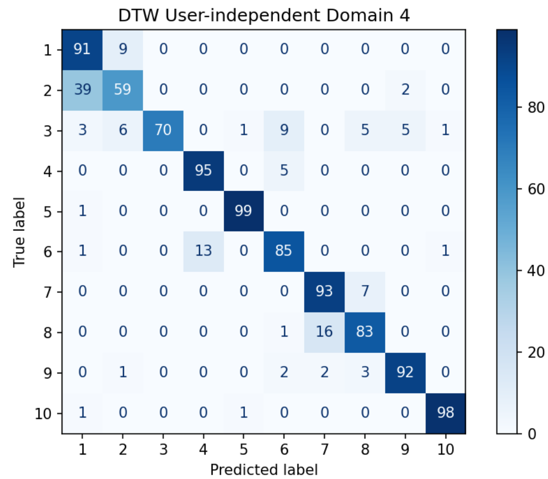

Figure 5: *Confusion matrix for DTW on Domain 4 (User-independent).*

IV-E Overfitting Analysis

Table 7 presents train-test gaps for parametric classifiers (RF, DT, LR). DTW, Edit Distance, and $1 are excluded because their training accuracy is 1.0 by construction (1-NN).

Domain	Protocol	DT (gap)	RF (gap)	LR (gap)
1	UI	**+0.128**	+0.080	+0.086
	UD	**+0.110**	+0.038	+0.033
4	UI	**+0.135**	**+0.124**	**+0.106**
	UD	**+0.106**	+0.023	+0.029

Table 7: *Train-test gaps (train_acc − test_acc) for parametric classifiers. Values > 0.10 in bold.*

Decision Tree presents the highest overfitting across all scenarios (gaps $\in [+0.106, +0.135]$), confirming its strong intrinsic variance [13], [15]. Despite hyperparameter selection via GridSearchCV (`max_depth`, `min_samples_split`, `min_samples_leaf`), a single tree remains sensitive to training set particularities, capturing specific patterns that do not generalise.

Random Forest significantly reduces overfitting compared to DT thanks to bagging [8]. In UI, gaps remain high (D1: +0.080, D4: +0.124), suggesting that even aggregation of 200 trees does not suffice to completely eliminate variance on complex inter-user gestures. In UD, gaps drop drastically (D1: +0.038, D4: +0.023), confirming that RF generalises excellently intra-user.

Logistic Regression presents the lowest gaps (∼0.03–0.09), consistent with its nature as a regularised linear model (L2 penalty via hyperparameter C) [16], [17]. LR is less subject to overfitting than RF or DT, but its weak capacity (linearity) limits its raw performance (0.851 on D4 UI, against 0.864 for RF).

Summary. Random Forest bagging effectively reduces variance compared to Decision Tree, justifying its adoption as a reference method for feature-based classifiers. Logistic Regression, although less flexible, offers interesting robustness to overfitting, at the cost of slightly lower performance.

IV-F Statistical Tests (RQ3)

Paired Wilcoxon signed-rank tests, corrected by Benjamini-Hochberg FDR at 5%, evaluate the significance of performance differences between methods in UI protocol.

IV-F1 Domain 1

Table 8 synthesises p-values. The \$1 recognizer is significantly superior to all other methods ($p_{\text{BH}} < 0.05$ for all 5 comparisons). Other notable results: (1) DTW vs Edit Distance: $p_{\text{BH}} = 0.0029$ (significant). DTW (0.918) surpasses Edit Distance (0.869) in a statistically robust manner. (2) DTW vs DT: $p_{\text{BH}} = 0.0013$ (significant). DTW surpasses DT (0.827). (3) DT vs RF: $p_{\text{BH}} < 0.001$ (significant). RF (0.912) clearly surpasses DT. (4) RF vs LR: $p_{\text{BH}} = 0.0264$ (significant). RF marginally surpasses LR (0.870).

Non-significant comparisons include: Edit Distance vs DT ($p_{\text{BH}} = 0.094$), Edit Distance vs RF ($p_{\text{BH}} = 0.297$), Edit Distance vs LR ($p_{\text{BH}} = 0.741$), DTW vs RF ($p_{\text{BH}} = 0.348$), DTW vs LR ($p_{\text{BH}} = 0.050$), and DT vs LR ($p_{\text{BH}} = 0.072$). These results suggest that, although \$1 clearly dominates, other methods form a "plateau" of similar performance (∼0.82–0.92), without statistically robust differences.

Method (A)	Edit Dist.	DTW	DT	RF	LR	\$1
Edit Distance	–	0.0029	0.0943	0.2966	0.7411	0.0000
DTW	**0.0029**	–	**0.0013**	0.3475	0.0502	0.0018
DT	0.0943	0.0013	–	0.0000	0.0718	0.0000
RF	0.2966	0.3475	**0.0000**	–	**0.0264**	0.0000
LR	0.7411	0.0502	0.0718	0.0264	–	0.0000
\$1	**0.0000**	**0.0018**	**0.0000**	**0.0000**	**0.0000**	–

Table 8: *Matrix of adjusted p-values (Benjamini-Hochberg correction, UI protocol). Bold values indicate that the row method is significantly superior to the column method ($\alpha = 0.05$).*

Statistically significant pairwise superiorities are: (i) \$1 > all methods (5/5), (ii) DTW > Edit Distance and DT (2/5), and (iii) RF > DT and LR (2/5).

IV-F2 Domain 4

Table 9 presents the p-values for Domain 4. Tests reveal an almost total absence of significance. Out of 15 comparisons, only DT vs RF is significant ($p_{\text{BH}} = 0.0064$), RF (0.864) surpassing DT (0.813). All other comparisons are non-significant. This result is explained by: (1) Performances are grouped in a narrow range (0.813–0.865, gap of 5.2 points). (2) Standard deviations are high (∼0.07–0.10), reflecting strong inter-user variability on complex 3D gestures.

Method (A)	Edit Dist.	DTW	DT	RF	LR	\$1
Edit Distance	–	0.6313	0.6313	0.6313	0.7744	0.6603
DTW	0.6313	–	0.2498	0.9261	0.7178	0.2552
DT	0.6313	0.2498	–	0.0064	0.2552	0.8736
RF	0.6313	0.9261	**0.0064**	–	0.8583	0.4466
LR	0.7744	0.7178	0.2552	0.8583	–	0.4466
\$1	0.6603	0.2552	0.8736	0.4466	0.4466	–

Table 9: *Matrix of adjusted p-values (Benjamini-Hochberg correction, UI protocol, Domain 4). Bold value indicates that the row method is significantly superior to the column method ($\alpha = 0.05$).*

Answer to RQ3: Domain 1: The \$1 recognizer is significantly superior to all other methods. DTW is significantly superior to edit distance and DT. RF is significantly superior to DT and LR. Other differences are not statistically robust. **Domain 4:** No method is significantly superior to others (except RF > DT). This absence of significance reflects strong inter-user variability and geometric complexity of the domain.

V Conclusion

This comparative study evaluated six 3D hand gesture recognition methods across two distinct domains using rigorous cross-validation protocols.

RQ1 (Preprocessing impact): Standardisation drastically improves performance for DTW, Edit Distance, and feature-based classifiers (gains up to +37.9 points on Domain 4), effectively neutralising inter-user scale variations. PCA denoising systematically degrades performance on both domains, revealing that PC3 encodes discriminant geometric information even when weak (5.8% variance on Domain 1). The \$1 recognizer performs best without external preprocessing thanks to its robust internal normalisation pipeline.

RQ2 (Best methods): On Domain 1 (quasi-planar gestures), the \$1 recognizer dominates in both UI (0.992) and UD (0.999), benefiting from its geometric normalisation optimised for planar trajectories. On Domain 4 (volumetric 3D gestures), DTW achieves best performance in UI (0.865) and UD (0.987), demonstrating superior robustness to complex spatial patterns through temporal alignment.

RQ3 (Statistical significance): On Domain 1, the \$1 recognizer is significantly superior to all methods (Wilcoxon + BH, $\alpha = 0.05$). DTW is significantly superior to edit distance and DT. RF is significantly superior to DT and LR. On Domain 4, only RF > DT is significant; other differences reflect strong inter-user variability on complex 3D gestures rather than fundamental method limitations.

RQ4 (UD improvement): The UD protocol systematically improves performance: moderate gains on Domain 1 (+0.7 to +10.3 points) where digits are already well discriminable, and massive gains on Domain 4 (+7.0 to +15.2 points), confirming that inter-user variability is the major bottleneck for volumetric gesture recognition.

References

[1] J. Huang, P. Jaiswal, and R. Rai, "Gesture-based system for next generation natural and intuitive interfaces," *Artificial Intelligence for Engineering Design, Analysis and Manufacturing*, vol. 33, no. 1, pp. 54–68, 2019.

[2] J. O. Wobbrock, A. D. Wilson, and Y. Li, "Gestures without libraries, toolkits or training: a $1 recognizer for user interface prototypes," in *Proc. 20th Annual ACM Symposium on User Interface Software and Technology (UIST)*, 2007, pp. 159–168.

[3] S. Theodoridis and K. Koutroumbas, *Pattern Recognition*, 4th ed. Academic Press, 2009.

[4] M. P. Deisenroth, A. A. Faisal, and C. S. Ong, *Mathematics for Machine Learning*. Cambridge University Press, 2020.

[5] J. Barbič, A. Safonova, J.-Y. Pan, C. Faloutsos, J. K. Hodgins, and N. S. Pollard, "Segmenting motion capture data into distinct behaviors," in *Proc. 17th Annual Conf. on Computer Animation and Social Agents (CASA)*, vol. 1, 2004, pp. 185–194.

[6] L. R. Rabiner and B.-H. Juang, *Fundamentals of Speech Recognition*. Prentice Hall, 1993.

[7] S. Garcia-Diez, C. Fernandez-Lozano, J. Dorado, and A. Pazos, "Feature extraction by using edit distance," *Expert Systems with Applications*, vol. 38, no. 4, pp. 3825–3830, 2011.

[8] L. Breiman, "Random forests," *Machine Learning*, vol. 45, no. 1, pp. 5–32, 2001.

[9] C. Ambroise and G. J. McLachlan, "Selection bias in gene extraction on the basis of microarray gene-expression data," *Proceedings of the National Academy of Sciences*, vol. 99, no. 10, pp. 6562–6566, 2002.

[10] L. Yu and H. Liu, "Efficient feature selection via analysis of relevance and redundancy," *Journal of Machine Learning Research*, vol. 5, pp. 1205–1224, 2004.

[11] C. Strobl, A.-L. Boulesteix, A. Zeileis, and T. Hothorn, "Bias in random forest variable importance measures: Illustrations, sources and a solution," *BMC Bioinformatics*, vol. 8, no. 25, 2007.

[12] S. Varma and R. Simon, "Bias in error estimation when using cross-validation for model selection," *BMC Bioinformatics*, vol. 7, no. 91, 2006.

[13] J. R. Quinlan, "Induction of decision trees," *Machine Learning*, vol. 1, no. 1, pp. 81–106, 1986.

[14] L. Breiman, J. Friedman, R. Olshen, and C. Stone, *Classification and Regression Trees (CART)*. Wadsworth, 1984.

[15] T. Hastie, R. Tibshirani, and J. Friedman, *The Elements of Statistical Learning: Data Mining, Inference, and Prediction*, 2nd ed. Springer, 2009.

[16] D. R. Cox, "The regression analysis of binary sequences," *Journal of the Royal Statistical Society: Series B*, vol. 20, no. 2, pp. 215–242, 1958.

[17] D. W. Hosmer, S. Lemeshow, and R. X. Sturdivant, *Applied Logistic Regression*, 3rd ed. Wiley, 2013.

[18] S. Kratz and M. Rohs, "A $3 gesture recognizer: Simple gesture recognition for devices equipped with inertial sensors," in *Proc. 15th International Conference on Intelligent User Interfaces (IUI)*, 2010, pp. 341–344.

[19] A. Mezari and I. Maglogiannis, "An easily customized gesture recognizer for assisted living using commodity mobile devices," *Journal of Healthcare Engineering*, vol. 2018, Article ID 3180652, 2018.

[20] J. Liu, L. Zhong, J. Wickramasuriya, and V. Vasudevan, "uWave: Accelerometer-based personalized gesture recognition and its applications," *Pervasive and Mobile Computing*, vol. 5, no. 6, pp. 657–675, 2009.

[21] S. Mitra and T. Acharya, "Gesture recognition: A survey," *IEEE Transactions on Systems, Man, and Cybernetics—Part C: Applications and Reviews*, vol. 37, no. 3, pp. 311–324, 2007.

[22] Y. Linde, A. Buzo, and R. Gray, "An algorithm for vector quantizer design," *IEEE Transactions on Communications*, vol. 28, no. 1, pp. 84–95, 1980.

[23] Y. Benjamini and Y. Hochberg, "Controlling the false discovery rate: A practical and powerful approach to multiple testing," *Journal of the Royal Statistical Society: Series B*, vol. 57, no. 1, pp. 289–300, 1995.

Appendix

Figures 6 and 7 present the box plots of sequence lengths per gesture class for domain 1 and domain 4, respectively. These plots motivate the use of alignment-robust methods such as DTW and edit distance, as discussed in Section III.

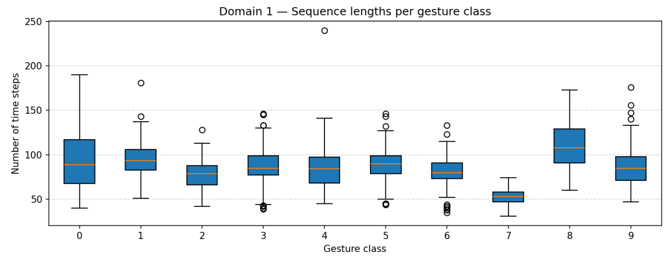

Figure 6: *Box plots of sequence lengths per gesture class for Domain 1 (Arabic numerals 0–9).*

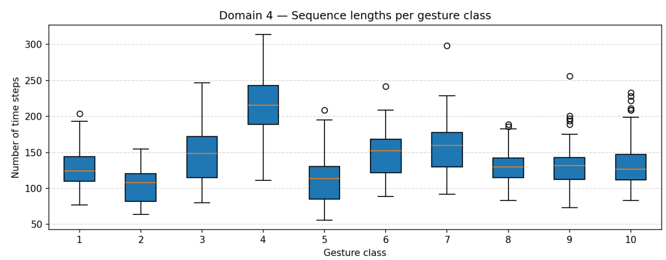

Figure 7: *Box plots of sequence lengths per gesture class for Domain 4 (CAD symbols).*

Figures 8 and 9 present the validation curves for the optimal number of k-means centroids (K_CLUSTERS) used in Edit Distance vector quantisation. These curves were obtained on standardised data in user-independent cross-validation. The selected values ($k = 20$ for Domain 1, $k = 15$ for Domain 4) correspond to the points where accuracy plateaus.

Figure 8: *Validation curve for K_CLUSTERS on Domain 1 (standardised data). The optimal value $k = 20$ is indicated by the vertical dashed line.*

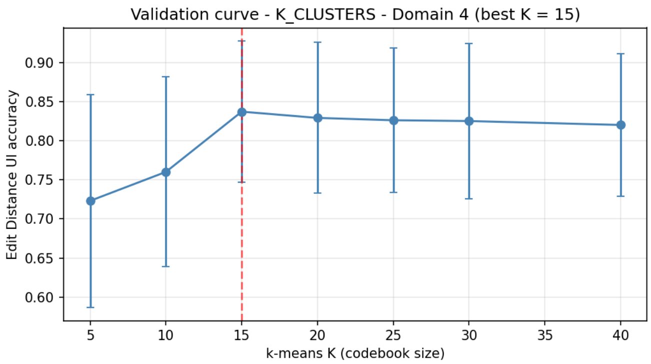

Figure 9: *Validation curve for K_CLUSTERS on Domain 4 (standardised data). The optimal value $k = 15$ is indicated by the vertical dashed line.*

Figures 10 and 11 present the validation curves for the k-NN parameter used in DTW, Edit Distance, and $1 recognizer classification. These curves were obtained in user-independent cross-validation. Although Domain 4 shows a marginal optimum at $k = 3$ (0.867), the negligible gain over $k = 1$ (0.865, +0.002) and standard practice in gesture recognition literature justify enforcing $k = 1$ across both domains.

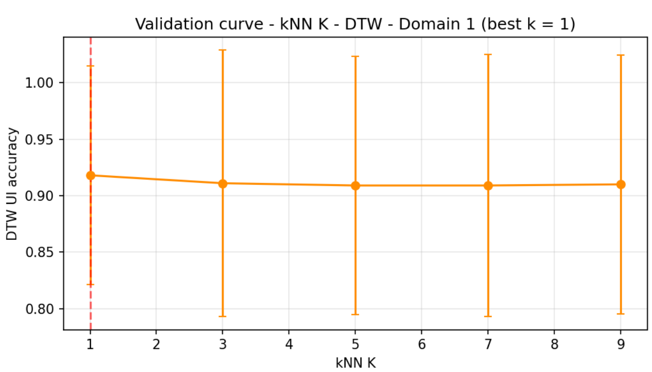

Figure 10: *Validation curve for k-NN on Domain 1 (DTW, UI). The enforced value $k = 1$ is indicated by the vertical dashed line and corresponds to the optimal performance.*

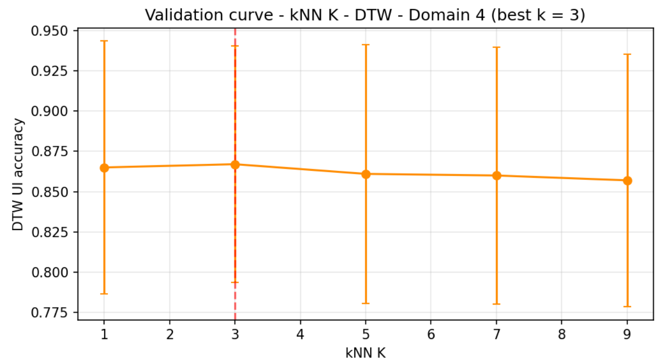

Figure 11: *Validation curve for k-NN on Domain 4 (DTW, UI). The enforced value $k = 1$ is indicated by the vertical dashed line, with a marginal difference to the curve optimum at $k = 3$.*